TECHNICAL REPORT

Biomedical Signal Processing & Topological State-Space Analysis

Takens' Delay Embedding System

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This document is self-contained. No prior reading is assumed beyond undergraduate-level linear algebra and basic signal processing.

You could find the code in the next link:
https://github.com/RoshTzsche/signals_attractor

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1. Introduction and Motivation

1.1 The Problem of Dimensionality in Neural Signals

The electroencephalogram (EEG) records voltage fluctuations at the scalp surface produced by the synchronised postsynaptic potentials of cortical pyramidal neurons. Each electrode channel captures a one-dimensional time series — a scalar observation of an underlying dynamical system whose true state space is high-dimensional and largely inaccessible to direct measurement.

A naive treatment of this signal as a stationary stochastic process ignores the rich geometric structure embedded within the temporal evolution of the data. The central challenge, therefore, is to reconstruct a faithful geometric representation of the underlying attractor from these partial, scalar observations.

This is not merely an academic exercise. The attractor geometry changes measurably during epileptic seizures, sleep stage transitions, anesthetic induction, and cognitive tasks. Being able to visualise and quantify those changes in near real-time has direct clinical and research utility.

1.2 What This Software Does

The Takens Attractor Dashboard is an interactive web application that ingests multi-format neurophysiological signal files and performs, in a single pipeline:

- Universal file ingestion supporting EDF, BDF, FIF, SET (EEGLAB), CSV/TXT/TSV, and NumPy .npy formats.
- Zero-phase FIR band-pass filtering to suppress DC drift and high-frequency noise.
- Autocorrelation-based optimal delay (τ) estimation via the Wiener-Khinchin theorem.
- Delay-coordinate embedding of the scalar time series into a three-dimensional state space (Takens' theorem).
- Spectral energy computation via the Complex Morlet Continuous Wavelet Transform (CWT) or Short-Time Fourier Transform (STFT) fallback, used as a colour gradient on the attractor.
- Welch power spectral density estimation and relative band-power decomposition across the five canonical EEG frequency bands.
- Hurst exponent, kurtosis, SNR, RMS, and artifact-rate quality metrics.
- An interactive Dash/Plotly web interface with drag-and-drop upload, epoch selector, parameter sliders, and WebGL-accelerated 3-D rendering.

1.3 Why This Approach Is Justified

Classical frequency-domain methods (FFT, band-pass spectrograms) treat the EEG as a superposition of sinusoidal components. This model is statistically convenient but physically incorrect: neural oscillations are non-stationary, non-linear, and non-Gaussian. Takens' embedding approach makes no linearity assumption; it works directly with the trajectory geometry of the underlying attractor, making it a principled alternative or complement to Fourier-based methods.

Furthermore, the attractor reconstruction provides visual diagnostic power that a power spectrum does not — a clinician can immediately perceive changes in orbit topology (e.g., the collapse of a complex torus to a near-fixed-point during burst suppression) without statistical hypothesis testing.

2. Mathematical Foundations

2.1 Dynamical Systems and Attractors

Consider a deterministic autonomous dynamical system governed by:

$$dx/dt = F(x), \quad x \in \mathbb{R}^n$$

where $F: \mathbb{R}^n \to \mathbb{R}^n$ is a smooth vector field. The solution x(t) traces a trajectory in phase space $\mathbb{R}^n$. As $t \to \infty$, dissipative systems contract phase-space volume and the trajectory settles onto a compact invariant subset $A \subset \mathbb{R}^n$ called the attractor.

For a harmonic oscillator, A is a closed orbit (limit cycle). For the Lorenz system, A is a fractal object of dimension ≈ 2.06. For resting EEG, the attractor dimension estimates typically fall between 2 and 8 depending on the recording site and mental state, indicating a low-dimensional but genuinely nonlinear process.

The key point: the attractor contains the 'essential geometry' of the system. If we can reconstruct A from observations, we can characterise the system without ever knowing F explicitly.

2.2 Takens' Delay Embedding Theorem

Suppose we can only measure a single scalar observable of the system:

$$s(t) = h(x(t)), \quad h: \mathbb{R}^n \to \mathbb{R}$$

Takens (1981) proved the following fundamental result:

> **NOTE:** THEOREM (Takens, 1981) — Let A be a compact manifold of dimension d and let s(t) be a generic smooth observable. For an embedding dimension m > 2d and a generic delay τ > 0, the delay map

$$\Phi_\tau : x(t) \mapsto [s(t), s(t+\tau), s(t+2\tau), \ldots, s(t+(m-1)\tau)]$$

is a smooth embedding of A into $\mathbb{R}^m$. That is, the delay-coordinate vector Φ_τ(x(t)) is topologically equivalent (diffeomorphic) to x(t).

In plain language: even though we only observe a single number at each moment, stacking m time-delayed copies of that number reconstructs a space that has the same topological structure — the same knots, loops, spirals, and holes — as the original attractor. This is a profound result because it means we lose no topological information by reducing to a scalar observable.

The implications for EEG are direct: one electrode channel is, in principle, sufficient to reconstruct the geometry of the underlying cortical dynamical attractor, provided we choose m and τ correctly.

> **PRACTICAL:** The theorem is generic, meaning the stated conditions hold for almost all smooth h and almost all τ. In practice, poor choices of τ (too small: coordinates are redundant; too large: they become statistically independent) degrade the reconstruction, so optimal estimation is critical.

2.3 Choosing the Embedding Dimension m

The theorem guarantees correctness for m > 2d where d is the attractor dimension. Since d is unknown a priori, the standard method is Cao's False Nearest Neighbours (FNN) algorithm or the E1/E2 criterion, which estimates the minimum m at which points that appear adjacent in (m)-dimensional space are no longer neighbours when projected back from (m+1)-dimensional space (i.e., they were 'false' neighbours due to projection artefacts).

The dashboard defaults to m = 3 (rendering always three dimensions for visualisation). For clinical analysis, m should be determined algorithmically. The slider allows m ∈ {3, 4, 5}, which covers most EEG scenarios (attractor dimension rarely exceeds 5 in clean recordings).

2.4 Optimal Lag Selection via Autocorrelation

The delay τ must be chosen such that consecutive embedding coordinates carry independent information. Two standard criteria exist:

2.4.1 First Zero of the Autocorrelation Function

The autocorrelation function R_xx(τ) is defined as:

$$R_xx(\tau) = E[(x(t) - \mu)(x(t+\tau) - \mu)] / \sigma^2$$

The first zero-crossing of R_xx identifies the smallest τ at which s(t) and s(t+τ) are linearly uncorrelated. This is the criterion implemented in the dashboard via the Wiener-Khinchin theorem, which states:

$$R_xx(\tau) = \mathcal{F}^{-1}\{S_xx(f)\}$$

where S_xx(f) is the power spectral density of x. Computing R_xx via FFT yields O(N log N) complexity rather than O(N²) for direct computation.

```
acf = signal.correlate(x_centered, x_centered, mode='full', method='fft')
```

The implementation computes the full autocorrelation, takes the positive-lag half, normalises to R_xx(0) = 1, and finds the first index where the sign changes (zero-crossing).

2.4.2 First Minimum of Average Mutual Information (AMI)

A theoretically superior criterion uses the Average Mutual Information I(τ):

$$I(\tau) = \Sigma\, p(s(t), s(t+\tau)) \cdot \log_2[p(s(t), s(t+\tau)) / (p(s(t))\cdot p(s(t+\tau)))]$$

This captures nonlinear dependencies that the linear autocorrelation misses. The first minimum of I(τ) identifies the τ at which knowledge of s(t) provides the least information about s(t+τ), ensuring maximum information richness in the embedding. This criterion (Fraser & Swinney, 1986) is not implemented in the current version but is a recommended extension.

2.5 Delay Embedding via NumPy Stride Tricks

The naive implementation of the delay map allocates an (M × m) matrix by copying data. For a 10-minute EEG at 256 Hz, this is 153,600 samples — resulting in a 3 × 153,600 matrix (~3.5 MB) with full data duplication.

The stride-tricks implementation avoids this by exploiting the memory layout of NumPy arrays. Instead of copying, it creates a new array descriptor (a view) that reads from the same memory buffer using custom strides:

```
shape   = (M, dim)
strides = (itemsize, itemsize * tau)    # advance 1 element per row, tau elements per
column
E = as_strided(x, shape=shape, strides=strides)
```

The result is a zero-copy O(1) memory operation. The (i, j) element of E reads x[i + j*τ] directly from the original buffer. This is correct because the buffer is contiguous and we are only reading it — no writes occur.

> **SAFETY:** as_strided is unsafe if the resulting view goes out of bounds. The code guards this with M = N - (dim-1)*τ > 0, which ensures the last element E[M-1, dim-1] = x[M-1 + (dim-1)*τ] = x[N-1] is within bounds.

2.6 Spectral Energy via Continuous Wavelet Transform

Standard FFT is computed over the entire signal and loses temporal information. The signal's instantaneous spectral content is captured by the Continuous Wavelet Transform:

$$(Wf)(a, b) = (1/\sqrt{a}) \cdot \int f(t) \cdot \psi^*((t-b)/a)\, dt$$

where ψ is the mother wavelet, a is the scale (inversely proportional to frequency), and b is the temporal position. The Complex Morlet wavelet:

$$\psi(t) = (\pi\, f_b)^{-1/2} \cdot \exp(2\pi i\, f_c\, t) \cdot \exp(-t^2/f_b)$$

is chosen because it optimises the time-frequency uncertainty product (Heisenberg-Gabor inequality) and has a well-defined centre frequency f_c, making the frequency-to-scale mapping intuitive.

The instantaneous spectral energy is the L^2 norm of the CWT matrix across all frequency scales at each time point:

$$E(t) = \Sigma_a\, |Wf(a,t)|^2$$

After min-max normalisation, E(t) is used as the colour gradient along the 3-D attractor curve, visually encoding whether the system was in a high-energy (broadband) or low-energy (narrow-band) state at each point of the trajectory.

If PyWavelets is unavailable, the dashboard falls back to the Short-Time Fourier Transform energy, which uses a sliding window of length N_seg:

$$STFT(t, f) = \int x(\tau) \cdot w(\tau-t) \cdot e^{-2\pi i f \tau}\, d\tau$$

$$E_STFT(t) = \Sigma_f\, |STFT(t, f)|^2$$

2.7 Welch Power Spectral Density

The Welch method (1967) reduces variance in PSD estimation by segmenting the signal into K overlapping windows of length L, computing the periodogram of each, and averaging:

$$\hat{S}_xx(f) = (1/K) \cdot \Sigma_\{k=1\}^\{K\} |\mathcal{F}\{w(t) \cdot x_k(t)\}|^2 \,/\, (fs \cdot \Sigma\, w^2(t))$$

where w(t) is the Hann window. The normalisation ensures Parseval's theorem holds: $\int \hat{S}_xx(f)\, df = \sigma^2$. Relative band powers are computed as:

$$P_band = (\int_\{f_lo\}^\{f_hi\} \hat{S}_xx(f)\, df) \,/\, (\int_0^\{fs/2\} \hat{S}_xx(f)\, df) \times 100\%$$

using the trapezoidal rule for numerical integration.

2.8 Hurst Exponent

The Hurst exponent H characterises the long-range dependence and self-similarity of a time series. It is estimated via the Rescaled Range (R/S) analysis:

$$E[R(n)/S(n)] \approx C \cdot n\text{^}H$$

where R(n) is the range of the cumulative deviation from the mean over a window of length n, and S(n) is the standard deviation. Taking logarithms:

$$\log(R(n)/S(n)) = H \cdot \log(n) + \log(C)$$

H is estimated by linear regression of log(R/S) on log(n). Interpretation:

- $H > 0.5$ — Persistent process: large values tend to be followed by large values (long-range correlation). Common in alpha-band EEG.
- $H \approx 0.5$ — Brownian motion (random walk): no long-range memory. Consistent with white noise.
- $H < 0.5$ — Anti-persistent process: large values tend to be followed by small values (mean-reverting). Seen in some pathological states.

EEG typically exhibits $H \in [0.55, 0.75]$ during normal waking rest, indicating mild long-range persistence. Values outside this range may warrant clinical attention.

3. Software Architecture and Code Walkthrough

3.1 Module Structure

The application is organised as a single-file Dash application (takens_attractor_dashboard.py) for portability, divided into six logical sections:

Section	Name	Lines (approx.)	Responsibility
1	Constants & Band Definitions	20	Global parameters, EEG frequency bands, visual theme
2	Multi-Format File Ingestion	90	Parsing EDF/BDF/FIF/SET/CSV/NPY into a unified channel dict
3	Signal Analysis Utilities	110	Tau estimation, phase space, PSD, Hurst, artifact metrics
4	Figure Builders	100	Plotly figure constructors for each visualisation panel
5	Dash Application Layout	130	Component tree: sidebar, dropzone, sliders, plots, cards
6	Callbacks	180	Reactive logic connecting UI state to analysis pipeline

Table 3.1. High-level module decomposition of the dashboard application.

3.2 Section 1: Constants and EEG Band Definitions

The five canonical EEG frequency bands are defined as a Python dictionary mapping band name to a tuple of (low_freq_Hz, high_freq_Hz, hex_colour):

```
EEG_BANDS = {
    "δ Delta":  (0.5,   4.0,  "#4361ee"),
    "θ Theta":  (4.0,   8.0,  "#7209b7"),
    "α Alpha":  (8.0,  13.0,  "#f72585"),
    "β Beta":   (13.0, 30.0,  "#fb8500"),
    "γ Gamma":  (30.0, 45.0,  "#06d6a0"),
}
```

The upper gamma boundary is set at 45 Hz to stay below the 50/60 Hz power-line interference notch, which most EEG systems filter. The delta lower boundary of 0.5 Hz (rather than 0 Hz) excludes DC drift and movement artefacts.

3.3 Section 2: Multi-Format File Ingestion

The ingest_file function accepts raw bytes and a filename string, routing to a format-specific parser based on the file extension. The return contract is uniform across all formats: a dictionary {channel_name: np.ndarray}, a sampling frequency float, and a format tag string.

3.3.1 EDF/BDF/FIF/SET Formats

These formats are read via MNE-Python (mne.io.read_raw_*). The file bytes are written to a NamedTemporaryFile, passed to MNE's reader, and the temp file is deleted. Up to 32 channels are

extracted to prevent memory overflow on dense-array recordings. A 0.5 Hz high-pass filter is applied to remove DC component and slow drift artefacts.

3.3.2 CSV/TXT/TSV Formats

Comma-, tab-, and space-delimited text files are read with pandas.read_csv. Non-numeric columns are coerced to NaN and dropped. A heuristic detects the time column by checking whether the column is strictly monotone-increasing with a coefficient of variation of less than 5% in its first differences — this correctly identifies both uniformly-sampled time vectors and near-uniform ones with floating-point jitter.

Sampling frequency is inferred from the median inter-sample interval of the time column. If no time column is found, fs defaults to 256 Hz (the most common clinical EEG sample rate).

3.3.3 NumPy .npy Format

Binary NumPy arrays are loaded with np.load(..., allow_pickle=False) for security (pickle-based .npy files can execute arbitrary code). A 1D array is treated as a single channel; a 2D array is assumed to be (samples × channels) with automatic transposition if the column count exceeds the row count, which is the correct heuristic for high-channel-count recordings stored in (channels × samples) layout.

3.4 Section 3: Signal Analysis Utilities

3.4.1 Zero-Phase Band-Pass Filter (bandpass)

A fourth-order Butterworth IIR filter is designed in the normalised frequency domain and applied with scipy.signal.filtfilt, which performs forward-backward filtering. Forward-backward application ensures zero phase distortion — the filtered signal has no time delay relative to the original, which is critical for preserving temporal relationships in the embedding.

$$H(j\Omega) = 1 / \sqrt{(1 + (\Omega/\Omega_c)^{2N})} \quad \text{where } N = 4$$

> **NUMERICAL:** IIR filters of order ≥ 4 can exhibit numerical instability in fixed-point arithmetic. The code operates in float64 throughout, making this a non-issue in practice. However, for very narrow bands or extreme cutoffs, second-order sections (sos) design would be more numerically stable.

3.4.2 Band Power Computation (compute_band_powers)

Welch PSD is computed with a window length of min(N, 4·fs) samples (4 seconds, or the entire signal if shorter). This provides a frequency resolution of at least 0.25 Hz — sufficient to resolve the delta/theta boundary at 4 Hz with two frequency bins to spare.

Relative band powers are computed by numerical integration (np.trapz) over the Welch PSD within each band boundary, normalised by the total integrated power.

3.4.3 Spectral Energy Gradient (spectral_energy_gradient)

This function computes a per-sample spectral energy value for use as the colour gradient on the attractor plot. Two implementation paths exist:

- Primary (PyWavelets available): CWT with 30 logarithmically-spaced scales covering 1–45 Hz using the Complex Morlet wavelet. The L^2 energy is summed across scales at each time point.
- Fallback (PyWavelets absent): STFT with noverlap = nperseg − 1, creating a sample-by-sample energy estimate. The STFT output is resampled to match the signal length via linear interpolation.

3.4.4 Signal Quality Metrics (signal_quality)

Five metrics are computed on the epoch:

Metric	Formula / Method	Clinical Significance
RMS Amplitude	$\sqrt{(E[x^2])}$	Overall signal magnitude; very low RMS may indicate electrode disconnection
SNR (dB)	$10 \cdot \log_{10}(\sigma^2_x / \sigma^2_\Delta x)$	Ratio of signal variance to finite-difference variance; proxy for spectral flatness
Kurtosis	$E[(x-\mu)^4] / \sigma^4 - 3$	Excess kurtosis > 5 suggests spike artefacts; kurtosis ≈ 0 for Gaussian noise
Hurst Exponent	R/S regression slope	Long-range temporal correlation; $H \in (0.5, 1.0)$ indicates persistence
Artifact %	% samples with $\lvert z \rvert > 4$	Proportion of samples exceeding 4 standard deviations from the mean

Table 3.2. Signal quality metrics computed per epoch.

3.5 Section 4: Figure Builders

Each figure builder is a pure function accepting NumPy arrays and returning a Plotly Figure object. This separation ensures the analysis pipeline is decoupled from the rendering layer, making unit testing straightforward.

- fig_timeseries — Scatter with dual trace: a thin line for visual continuity and coloured markers for spectral energy. Hover shows time and amplitude.
- fig_phase_space — Scatter3d with a coloured line trace. WebGL rendering is used implicitly by Plotly for 3D plots, enabling smooth interaction even with 150,000+ points.
- fig_psd — Scatter with area fill and per-band vertical rectangles (vrect). Hover shows frequency and power in dB/Hz.
- fig_band_radar — Barpolar chart displaying the relative power in each band as a polar bar. Each band is assigned a distinct colour from the BAND_COLORS palette.

3.6 Section 5: Dash Application Layout

The layout uses a flexbox two-column structure: a fixed-width 280 px sidebar containing controls, and a fluid main area containing the visualisations. This is achieved without CSS frameworks by setting display: flex on the outer container.

The drag-and-drop upload is implemented with dcc.Upload, Dash's built-in upload component, which wraps the browser's HTML5 File API. The file content is delivered to the callback as a Base64-encoded data URI.

All user-adjustable parameters are exposed as Dash controls:

Control	Component	Range / Options	Effect
File upload	dcc.Upload (drop zone)	Any accepted extension	Triggers ingestion; populates channel dropdown
Channel	dcc.Dropdown	All channels in file	Selects which channel to analyse
Epoch window	dcc.RangeSlider	0 – max_duration (s)	Crops signal to selected time window
Embed. dim.	dcc.Slider	3, 4, 5	Dimension m of Takens embedding
Max τ search	dcc.Slider	10 – 500 ms	Upper bound for autocorrelation lag search

Table 3.3. Interactive control mapping.

3.7 Section 6: Callbacks

3.7.1 Upload Callback (on_upload)

Triggered by changes to the dcc.Upload contents property. The callback decodes the Base64 data URI, calls ingest_file, and populates: the channel dropdown options and default value; the epoch slider maximum; and a file-info badge showing format, channel count, sampling rate, and duration. Data is stored in dcc.Store components as JSON-serialisable Python dictionaries.

> **SCALABILITY:** Storing large recordings (>10 minutes, many channels) as JSON in dcc.Store can consume significant browser memory. Production deployments should store data server-side (e.g., using Redis or a temporary file cache) and pass only metadata to the client.

3.7.2 Analysis Callback (run_analysis)

Triggered by the Run button. The callback:

1. Retrieves the stored channel data and crops to the selected epoch.
2. Calls compute_optimal_tau to estimate τ via FFT autocorrelation.
3. Calls phase_space to construct the delay-embedding matrix E.
4. Calls spectral_energy_gradient to compute the colour gradient.
5. Calls compute_band_powers to compute Welch PSD and relative powers.
6. Calls signal_quality to compute the five quality metrics.
7. Calls the four figure builders and returns all outputs in a single callback.

The callback follows Dash's single-output-per-callback best practice where feasible, but returns all fourteen outputs simultaneously to avoid redundant computation. All exceptions are caught and surfaced to the user as a red status bar with the exception message.

4. Supported File Formats

A key design goal of the v2.0 dashboard is universal file ingestion. The following table summarises all supported formats, their origins, and parsing strategy.

Extension	Format Name	Origin / Standard	Parser	Notes
.edf	European Data Format	Kemp et al., 1992 (IEEE)	MNE read_raw_edf	Clinical gold standard; stores annotations, patient ID
.bdf	BioSemi Data Format	BioSemi BV extension of EDF	MNE read_raw_edf	24-bit samples; used by BioSemi ActiveTwo
.fif	Functional Imaging File	Neuromag/Elekta (MEG/EEG)	MNE read_raw_fif	Rich metadata; coordinate frames; projections
.set	EEGLAB Dataset	Delorme & Makeig, 2004	MNE read_raw_eeglab	Supports ICA decompositions and event markers
.csv/.tsv/.txt	Comma/Tab-Separated Values	Generic	pandas read_csv	Auto-detects time column and sampling rate
.npy	NumPy Binary Array	NumPy project	numpy.load	1D or 2D; allow_pickle=False for security

Table 4.1. Supported input file formats.

Formats NOT currently supported include: .mat (MATLAB workspace files, require scipy.io.loadmat and field-name guessing), .h5/.hdf5 (HDF5, no standard EEG schema), .mff (EGI Metafile Format), and .ns5 (Blackrock Microsystems). These can be added by implementing additional branches in the ingest_file function.

5. EEG Clinical Insights Engine

5.1 Frequency Band Interpretation

The dashboard automatically interprets the dominant frequency band and generates human-readable state hypotheses. The interpretation rules are evidence-based and documented in standard neurological references (Niedermeyer & Lopes da Silva, 2011):

Band	Frequency Range	Normal Association	Pathological / Altered Association
δ Delta	0.5 – 4 Hz	Deep sleep (NREM stage 3/4)	Focal: cortical lesion. Diffuse: encephalopathy, deep anaesthesia
θ Theta	4 – 8 Hz	Drowsiness, memory encoding (hippocampus)	Excess waking theta: ADHD, mild cognitive impairment, frontal lobe dysfunction
α Alpha	8 – 13 Hz	Relaxed wakefulness, eyes closed	Alpha slowing: hypothyroidism, dementia. Alpha coma: poor prognostic sign
β Beta	13 – 30 Hz	Active cognition, motor planning	Excess beta: benzodiazepine effect. Low beta: frontal hypoactivation
γ Gamma	30 – 45 Hz	Feature binding, high-frequency cognition	Excess gamma: some forms of epilepsy, psychosis

Table 5.1. EEG frequency band associations and clinical correlates.

5.2 Automated State Detection Logic

The insights panel generates state hints using threshold rules on the computed relative band powers. The current rules are:

- Alpha > 30% → 'Relaxed / eyes-closed state'
- Delta > 40% → 'Deep sleep or pathological slowing'
- Theta > 25% → 'Drowsiness or memory encoding'
- Kurtosis > 10 → 'High-amplitude spike artefacts likely'

These thresholds are empirically derived from the literature and should be treated as suggestions, not diagnoses. A full clinical-grade automated interpretation system would require a trained classifier (SVM, random forest, or deep neural network) on labelled EEG data.

5.3 Hurst Exponent and Brain State

The Hurst exponent computed via R/S analysis provides a summary statistic of the temporal scaling behaviour of the EEG. Several studies have reported characteristic H values for different brain states:

Brain State	Typical H Range	Source
Resting wakefulness (eyes open)	0.55 – 0.70	Watters (1998); Wijnants et al. (2012)
Resting wakefulness (eyes closed)	0.65 – 0.80	Stam & de Bruin (2004)
Active cognition / task	0.50 – 0.62	Linkenkaer-Hansen et al. (2001)
Slow-wave sleep	0.75 – 0.90	Bhattacharya et al. (2003)
Absence seizure	0.30 – 0.50	Srinivasan et al. (1998)

Table 5.2. Reported Hurst exponent ranges for canonical EEG states (indicative; values vary by channel and recording setup).

6. Historical Context and Related Work

6.1 The Origins of Delay Embedding in Neuroscience

The mathematical foundation for delay embedding was laid by Floris Takens in his 1981 paper 'Detecting Strange Attractors in Turbulence,' published in the Springer Lecture Notes in Mathematics (Vol. 898, pp. 366–381). The theorem was later strengthened by Sauer, Yorke, and Casdagli (1991) to handle fractal attractors and non-smooth observables.

The first application of Takens' theorem to EEG data was reported by Babloyantz et al. (1985) in the journal Physics Letters A, who computed attractor dimension estimates for sleep EEG and found values consistent with deterministic dynamics ($d \approx 5$–7 during slow-wave sleep). This was a landmark paper establishing that brain dynamics could be described by deterministic low-dimensional attractors rather than purely stochastic processes.

Subsequent influential work includes:

- Rapp et al. (1989) — Chaos in brain electrical activity during anesthesia induction, documented via correlation dimension (D2) estimates.
- Stam et al. (1999) — Applied nonlinear dynamics measures (including embedding) to differentiate EEG in Alzheimer's disease from healthy controls, finding reduced attractor complexity in Alzheimer's patients.
- Le Van Quyen et al. (2001) — Pre-ictal attractor changes in epilepsy, showing that the state-space trajectory changes character up to 30 minutes before a clinical seizure — a finding with major implications for seizure prediction.

6.2 Existing Software and Tools

Several software systems have implemented EEG attractor analysis prior to this dashboard:

Tool / Library	Year	Platform	Key Capability	Limitation vs. This Work
TISEAN	1998	C / Fortran CLI	Full nonlinear time series toolkit (FNN, Lyapunov, etc.)	No interactive UI; command-line only; no web deployment
BrainStorm	2011	MATLAB GUI	Full EEG/MEG pipeline including source modelling	Requires MATLAB licence; no attractor visualisation
MNE-Python	2013	Python library	Gold-standard EEG/MEG I/O, filtering, ICA	No nonlinear dynamics; no attractor rendering
pyEDFlib	2014	Python library	Lightweight EDF reader	I/O only; no analysis or visualisation
nolds	2016	Python library	Hurst, DFA, Lyapunov, correlation dimension	Analysis only; no UI; no attractor rendering

Tool / Library	Year	Platform	Key Capability	Limitation vs. This Work
This Dashboard	2026	Python/Dash web app	End-to-end pipeline: ingest → embed → visualise → insights	No FNN, no AMI, no Lyapunov exponents (yet)

Table 6.1. Comparison with related tools in the EEG analysis ecosystem.

The distinguishing contribution of this dashboard is the combination of: (a) multi-format universal ingestion, (b) automated optimal-τ estimation, (c) spectrally-coloured 3-D attractor rendering with interactive WebGL, and (d) clinical insight generation — all in a single zero-configuration web application.

6.3 What Has and Has Not Been Done Before

Phase-space reconstruction from EEG is a mature technique with a 40-year history. What is novel in this implementation is the integration layer:

- The combination of CWT-derived spectral energy as a colour gradient on the attractor is, to the authors' knowledge, not present in any prior open-source tool. It provides a direct visual mapping between the frequency content of the signal and the geometry of its attractor — connecting the two dominant EEG analysis paradigms in a single figure.
- The stride-tricks zero-copy embedding is an implementation detail rarely documented in the EEG literature, which typically uses full data copies. For long recordings this reduces peak memory by a factor equal to the embedding dimension.
- The Dash-based interactive web interface for Takens embedding is novel. Prior tools require MATLAB, R, or command-line Python. The drag-and-drop web interface lowers the barrier for clinical adoption significantly.

7. Strengths, Limitations, and Recommendations

7.1 Strengths

Dimension	Strength	Technical Justification
Mathematical rigour	Takens embedding is provably topologically faithful	Diffeomorphism guarantee; attractor geometry is preserved
Computational efficiency	O(N log N) autocorrelation; O(1) memory embedding	FFT convolution; stride-tricks zero-copy view
Zero-phase filtering	filtfilt eliminates group delay distortion	Forward-backward application cancels phase response
Format universality	Six file formats supported from a single entry point	Unified return contract; format-specific error messages
Robustness	All exceptions surfaced in UI; fallbacks for missing dependencies	try/except in every critical path; PyWavelets fallback to STFT
Interactivity	Parameter changes do not require script restarts	Dash reactive callbacks; all analysis re-runs on button press
Reproducibility	All parameters (τ, epoch, dim) visible and adjustable	No hidden defaults; τ is computed and displayed, not hard-coded

Table 7.1. Key strengths of the implementation.

7.2 Limitations and Known Weaknesses

Limitation	Impact	Recommended Fix
Autocorrelation τ (not AMI)	Misses nonlinear dependencies; τ may be suboptimal for highly non-Gaussian signals	Implement Fraser-Swinney AMI criterion
Fixed dimension m = 3	Embedding dimension theorem requires m > 2d; d may exceed 1 for multi-modal signals	Add Cao FNN algorithm for adaptive m selection
No stationarity test	Welch PSD and Hurst estimates assume stationarity; nonstationary segments give misleading results	Add KPSS or ADF stationarity test; flag if rejected
dcc.Store JSON serialisation	Large recordings bloat browser memory; multi-GB files will crash the tab	Implement server-side session storage with file paths
R/S Hurst estimate variance	The simplified R/S estimator has known upward bias for short time series (N < 1000)	Use DFA (Detrended Fluctuation Analysis) for H
No artefact rejection	Eye blinks, EMG, and electrode pops distort the attractor	Implement ICA-based artefact removal (MNE ICA API)
Single-channel analysis	Phase-space reconstruction from one channel ignores spatial EEG topology	Extend to multi-variate embedding or connectivity measures

Limitation	Impact	Recommended Fix
No Lyapunov exponent	Cannot quantify attractor stability or chaos quantitatively	Implement Kantz or Rosenstein algorithm for λ_1

Table 7.2. Known limitations and recommended extensions.

7.3 Security Considerations

- File parsing: numpy.load is called with allow_pickle=False, preventing arbitrary code execution via serialised Python objects in .npy files.
- MNE temp files: temporary EDF/BDF/FIF files are written to the system temp directory and deleted immediately after reading. A crash between write and delete leaves a temp file; production deployments should use a context manager.
- No authentication: the Dash server binds to localhost by default. Binding to 0.0.0.0 for network deployment without authentication would expose patient data. Use a reverse proxy with TLS and authentication for clinical deployments.
- HIPAA/GDPR: EEG files may contain patient-identifying information (EDF header fields). This application does not sanitise, log, or transmit this information, but operators must ensure compliance with applicable data protection regulations.

7.4 Performance Benchmarks

Approximate runtimes on a modern consumer CPU (Intel Core i7, single thread) for a 256 Hz, 15-second epoch (3,840 samples):

Pipeline Stage	Estimated Runtime	Dominant Operation
File ingestion (EDF)	~300 ms	MNE EDF parser + filter (C extension)
Optimal τ estimation	< 5 ms	FFT autocorrelation (O(N log N))
Phase space construction	< 1 ms	stride-tricks (O(1) memory mapping)
CWT spectral energy (PyWavelets)	~400 ms	30-scale CWT matrix computation
STFT spectral energy (fallback)	< 30 ms	Overlap-add STFT
Welch PSD + band powers	< 10 ms	scipy.signal.welch
Signal quality metrics	< 5 ms	R/S Hurst regression
Plotly figure construction	~50 ms	JSON serialisation of trace data

Table 7.3. Approximate per-stage runtimes for a 15-second epoch at 256 Hz (single thread, no GPU).

8. Use Cases

8.1 Research Applications

8.1.1 Attractor Geometry During Cognitive Tasks

Researchers studying the relationship between cognitive load and neural complexity can use the dashboard to visually compare attractor geometry across experimental conditions. The change from a low-complexity attractor (near-periodic, during repetitive motor tasks) to a high-complexity one (space-filling, during problem-solving) is immediately apparent in the 3-D view without requiring numerical correlation-dimension estimates.

8.1.2 Sleep Stage Classification

The five EEG bands and the Hurst exponent provide a rapid preliminary check of sleep stage transitions. During slow-wave sleep, delta band power exceeds 50%, the attractor collapses to a near-periodic low-dimensional orbit, and H increases toward 0.8. During REM sleep, the attractor expands and the alpha/beta mixture increases. These signatures can be used as a sanity check before automated sleep staging algorithms are applied.

8.1.3 Anaesthetic Depth Monitoring

Propofol and volatile anaesthetic agents induce characteristic EEG signatures: alpha-delta dissociation at sedative doses and burst suppression at surgical depths. The attractor provides a continuous visual indicator of anaesthetic depth — the transition from a complex waking attractor to a near-limit-cycle (deep anaesthesia) is one of the most dramatic topology changes observable in EEG.

8.2 Clinical Applications

8.2.1 Seizure Monitoring

Epileptic seizures are associated with a transient reduction in attractor dimensionality (hypersynchrony: many neurons firing together, reducing degrees of freedom). The dashboard can be used as a visual screening tool for ictal activity in retrospective recordings. The spectral energy gradient will show a dramatic increase in delta/theta power during the ictal phase, and the attractor will show a qualitative change in topology.

> **CLINICAL DISCLAIMER:** This tool is NOT a certified medical device and must NOT be used for primary clinical diagnosis. All clinical decisions must be made by qualified healthcare professionals using validated diagnostic tools.

8.2.2 Artefact Identification

Electrode disconnection, electrode pop, and EMG contamination produce characteristic signatures: very high kurtosis (>50), high artifact%, and attractor trajectories that consist of nearly vertical lines (impulsive

artefact) or dense noise balls (EMG). These are immediately visible in the 3-D view and flagged by the quality metrics panel.

8.3 Educational Applications

The dashboard is an effective teaching tool for courses on nonlinear dynamics, time series analysis, and biomedical signal processing. It allows students to interactively explore the effect of embedding parameters:

- Setting τ too small (e.g., τ = 1 sample): the attractor collapses onto the diagonal $x(t) = x(t+\tau)$, producing a thin streak along a 45° line.
- Setting τ too large (e.g., τ = 5 seconds): the coordinates become statistically independent and the attractor fills the embedding cube uniformly, losing all geometric structure.
- Setting optimal τ: the attractor unfolds into a recognisable curved manifold whose shape depends on the signal's dominant oscillation.

These three regimes can be explored live by adjusting the max-τ slider and re-running the analysis, making the abstract theory of Takens embedding concrete and intuitive.

9. Installation, Dependencies, and Deployment

9.1 Dependency Table

Package	Version (tested)	Purpose	Optional?
Python	3.9 – 3.12	Runtime	No
dash	≥ 2.14	Web application framework	No
dash-bootstrap-components	≥ 1.5	Bootstrap CSS for layout	No
plotly	≥ 5.18	Interactive figure library	No
numpy	≥ 1.24	Array operations, stride tricks	No
scipy	≥ 1.11	Signal filtering, Welch PSD, STFT	No
pandas	≥ 2.0	CSV parsing, numeric coercion	No
mne	≥ 1.5	EDF/BDF/FIF/SET file I/O	Yes (required for those formats)
pywavelets	≥ 1.5	Complex Morlet CWT for spectral energy	Yes (STFT fallback used if absent)

Table 9.1. Python dependencies. 'Optional' dependencies degrade gracefully if absent.

9.2 Installation Commands

Conda (recommended for scientific environments):

```
conda install -c conda-forge dash dash-bootstrap-components plotly numpy scipy pandas mne pywavelets
```

pip (faster; use inside a virtual environment):

```
pip install dash dash-bootstrap-components pywavelets mne
```

Minimum viable install (CSV/NPY only, no EDF support):

```
pip install dash dash-bootstrap-components
```

9.3 Running the Application

```
python takens_attractor_dashboard.py
```

Open a browser and navigate to http://127.0.0.1:8050. The server starts in production mode (debug=False) to suppress Dash's developer overlay.

9.4 Production Deployment

For network deployment, replace app.run() with a WSGI server:

```
gunicorn takens_attractor_dashboard:server -b 0.0.0.0:8050 -w 4
```

Use NGINX as a reverse proxy with TLS termination. Never expose raw Dash to the public internet without authentication, especially when processing patient data.

10. Recommended Future Extensions

The following extensions are prioritised by impact-to-implementation-effort ratio:

Priority	Extension	Impact	Estimated Effort
High	Average Mutual Information τ selection (Fraser-Swinney)	Correct τ for non-Gaussian signals; reduces embedding distortion	1–2 days
High	Cao's False Nearest Neighbours for adaptive m selection	Theoretically correct embedding dimension; prevents over-embedding	2–3 days
High	ICA-based artefact rejection (MNE FastICA)	Clean recordings → more accurate attractor geometry	3–5 days
Medium	Correlation dimension D_2 estimation (Grassberger-Procaccia)	Quantitative attractor complexity; clinical biomarker	2–3 days
Medium	Largest Lyapunov exponent λ_1 (Rosenstein algorithm)	Quantify chaos and attractor stability	2–4 days
Medium	Multi-channel joint embedding	Spatial topology; functional connectivity in phase space	5–7 days
Medium	Detrended Fluctuation Analysis (DFA) for Hurst exponent	Lower-bias H estimate; handles nonstationarity better than R/S	1–2 days
Low	Export to PDF / PNG	Clinical reporting integration	1 day
Low	Recurrence Plot and Recurrence Quantification Analysis (RQA)	Complements attractor view; quantifies recurrence patterns	3–4 days
Low	Server-side session storage (Redis/SQLite)	Handles multi-GB recordings without browser memory overflow	2–3 days

Table 10.1. Recommended future extensions in priority order.

11. References

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12. Glossary

Term	Definition
Attractor	The invariant set in phase space to which a dynamical system evolves asymptotically from nearby initial conditions.
CWT	Continuous Wavelet Transform. A time-frequency decomposition that uses scaled and translated copies of a mother wavelet to localise energy in both time and frequency.
Delay Embedding	The construction of a multi-dimensional vector from time-delayed samples of a scalar signal: [s(t), s(t+τ), s(t+2τ), …].
Diffeomorphism	A smooth bijective map with a smooth inverse. Takens' theorem guarantees the delay map is a diffeomorphism of the attractor.
EDF	European Data Format. A standard file format for biological signals, standardised by Kemp et al. (1992).
FIR / IIR	Finite / Infinite Impulse Response. Classes of digital filters with finite or infinite memory length.
filtfilt	Forward-backward filtering; applies a filter twice in opposite directions to achieve zero-phase response.
Hurst Exponent (H)	A statistical measure of long-range temporal correlation. H = 0.5 for random walk; H > 0.5 for persistent processes.
Kurtosis	The fourth standardised moment. Measures the 'tailedness' of a distribution. Gaussian kurtosis = 3; excess kurtosis = kurtosis − 3.
Phase Space	The abstract space in which all possible states of a dynamical system are represented, with each possible state corresponding to one unique point.
PSD	Power Spectral Density. The distribution of signal power per unit frequency, in units of V^2/Hz or equivalent.
SNR	Signal-to-Noise Ratio. Here estimated as the ratio of signal variance to finite-difference variance, in decibels.
Stride Tricks	A NumPy technique for creating array views with custom memory strides, enabling zero-copy array reshaping and windowing.
Tau (τ)	The embedding lag: the time delay between consecutive components of the delay-coordinate vector.
Welch Method	A variance-reduced PSD estimator using overlapping windowed periodogram averaging.
Wiener-Khinchin	Theorem stating that the autocorrelation function and power spectral density of a wide-sense stationary process are a Fourier transform pair.